Algebras and Representations

Twenty-one chapters have studied operators one at a time, or in commuting families. Chapter 9 asked for a basis in which a single operator is simple and found eigenvectors; Chapter 10 finished the question with the Jordan and rational forms; Chapters 11 to 13 answered it again with an inner product to respect. Then the question widened once, in Chapter 9 §09 and Chapter 12 §07: one basis good for a whole commuting family at a time. This chapter widens it once more, and to the last setting in which the question still has a clean answer — a set of operators closed under addition, under scalars, and under composition, with no assumption that its members commute. Such a set is an **algebra**, and the chapter's main theorem says that over an algebraically closed field there is essentially one building block, a full matrix algebra, and that everything semisimple is a product of those.

Nothing forced the earlier stopping point except habit. The polynomials in a fixed operator form an algebra; so do the circulants of Chapter 12 §09, the matrices commuting with a given matrix, the upper triangular matrices, the block diagonal matrices, and the linear symmetries of a figure. Every one of those the book has met and handled separately. What they have in common is the only thing this chapter needs, and naming it is most of the work.

The second half is the same statement read for groups. A group cannot be added, so linear algebra cannot reach it directly; but the formal combinations of group elements do form an algebra, the **group algebra** $F[G]$, and a group acting by invertible operators is exactly a representation of that algebra. Everything proved about algebras then applies word for word. Maschke's theorem says $F[G]$ is semisimple as soon as the characteristic of F does not divide $|G|$; the structure theorem splits it into matrix blocks; and the traces of those blocks — the **characters** — turn out to be an orthonormal basis of the functions that are constant on conjugacy classes. One square array of numbers then records every complex representation of the group, up to equivalence, and the number of rows is the number of conjugacy classes.

One debt is paid in full. For a finite abelian group every block has size one, the character table is a single unitary matrix, and for the cyclic group that matrix is Chapter 12 §09's Fourier matrix, entry for entry. The discrete Fourier transform, the convolution theorem, and "every circulant is a polynomial in the shift" were representation theory all along, and §09 proves each of the three identifications.

This is the last of the book's chapters of general theory. The chapter that follows is about arithmetic, algorithms, and applications, and nothing here depends on it.

23.-1.1 Conventions

- **Algebras are associative, unital, and finite-dimensional over** F, unless a statement says otherwise. The one standing exception is $F[x]$, kept as a source of counterexamples and always named as infinite-dimensional where it is used that way. Nothing requires commutativity, and nothing requires a nonzero element to have an inverse: an algebra is not a group, and §01 closes with a warning to that effect.
- **The definition is already in the book.** Chapter 14 §12's Definition 14.12.1 is used as it stands; §01 builds on it rather than restating it, adding Definition 23.1.1, Definition 23.1.2, Definition 23.1.3, and Definition 23.1.4.
- $\mathcal{L}(V)$ **and** $\mathrm{End}(V)$ **name the same object.** The book's default is $\mathcal{L}(V)$; §§05, 06, and 07 write $\mathrm{End}(V)$ where the algebra structure is the point. Likewise $\mathrm{End}_A(V) \subseteq \mathcal{L}(V)$, with the subscript never dropped, is the set of maps respecting an action of A.
- $F\langle S\rangle$ **is the subalgebra generated by** S and $\langle p\rangle$, without the F, is Chapter 6's ideal

generated by p. They are different and usually different sizes; §01 says so where the notation is introduced.

- S' **is the commutant**, the set of elements commuting with every member of S; $S'' = (S')'$ is the double commutant, and $Z(A) = A'$ is the center. The definition asks nothing of S — not that it be a subspace, not that its own members commute.
- **"Simple" is two words.** An **algebra** is simple when its only two-sided ideals are $\{0\}$ and itself; a **space** on which an algebra acts is simple, equivalently irreducible, when its only invariant subspaces are $\{0\}$ and itself. Neither implies the other, and §03 opens with the warning and the witness. **"Irreducible" is three words**, since Chapter 6 calls a polynomial irreducible (Definition 6.3.1) and Chapter 19 §02 calls a nonnegative matrix irreducible; §03's proof of Theorem 23.3.7 uses the first and the second within two lines of each other, and the argument is what tells them apart.
- **"Semisimple" is three words, and §03 reconciles them.** An algebra **acts semisimply on a space** when every invariant subspace has an invariant complement; an **algebra is semisimple** when it acts semisimply on itself, that is, when every left ideal has a complementary left ideal; and Chapter 10 §08 called an **operator** semisimple when its minimal polynomial is a product of distinct irreducibles. Theorem 23.3.7 proves the third is the first, applied to $F[T]$.
- $\operatorname{rad} A$ **is the radical of an algebra** (Definition 23.3.5), the set of elements killed by every irreducible representation. Chapter 14 §01's $\operatorname{rad}(\beta)$ is the radical of a bilinear form; the argument tells them apart and they never meet.
- $\mathbf{E}_{ij}$ **is the matrix unit** with a 1 in position (i, j), and δ_g is the basis vector of $F[G]$ attached to the group element g, so that $\delta_g \delta_h = \delta_{gh}$. The Kronecker delta always carries two numerical subscripts, δ_{jk}, and no group element is ever written as a pair of numbers.
- **Groups are finite from §07 on**, written multiplicatively. $\chi_\rho(g) = \operatorname{tr} \rho(g)$ is the character of a representation; only for an abelian group is a character a homomorphism, and §09 says so before adopting that reading and writing $\widehat{G}$ for the group of them.

23.-1.1.1 Which field, and where the hypothesis bites

Most theorems in this chapter are false over the wrong field, and it is worth saying once, exactly, which ones need what. §10 sets the same information out as a table, with a counterexample in every row.

Sections 1, 2, and 3 work over an arbitrary field F, with no hypothesis at all. That includes the chapter's first real theorem, Theorem 23.2.2, whose proof uses nothing about F beyond the existence of inverses of nonzero scalars; the classification of the left ideals of $M_n(F)$; the first isomorphism theorem for algebras; and Theorem 23.3.3, which is the section's engine and holds over $\mathbb{F}_2$ as readily as over $\mathbb{C}$.

Algebraic closure is asked for at exactly five places. The second half of Schur's lemma, Theorem 23.4.1 (b), and its three corollaries Corollary 23.4.3, Corollary 23.4.4, and Corollary 23.4.5; Burnside's theorem Theorem 23.5.2; and Wedderburn's structure theorem Theorem 23.6.1. In every one of them the hypothesis buys the same single thing — an operator on a nonzero space has an eigenvalue — and that is the only thing it buys. The **first** half of Schur's lemma needs nothing of the field and is used over arbitrary fields. **The double commutant theorem needs nothing of the field either**: Theorem 23.5.5 holds over every F, and what it asks instead is that A act semisimply and contain id_V. Both of those are shown by §05 to be necessary.

One matrix is the witness for every failure of algebraic closure. Rotation of $\mathbb{R}^2$ through $2\pi/n$ with $n \geq 3$ has no real eigenvalue. The algebra it generates is commutative, it acts irreducibly on $\mathbb{R}^2$, its commutant is a copy of $\mathbb{C}$ rather than $\mathbb{R}$, and it is a two-dimensional subalgebra of $M_2(\mathbb{R})$ and so is not all of $\operatorname{End}(\mathbb{R}^2)$. That one example breaks Schur's second

half, breaks the corollary that irreducible plus commutative means one-dimensional, and breaks Burnside. §04 computes the commutant in full.

A hypothesis on the characteristic enters only with groups, and only in one theorem. Theorem 23.7.3 needs $\operatorname{char} F \nmid |G|$, because its proof divides by $|G|$; the witness is $\mathbb{Z}/p\mathbb{Z}$ acting on $\mathbb{F}_p^2$ by unipotent matrices, where the invariant line has no invariant complement. Finiteness of G is used twice over — to average and to count — and §07 shows $\mathbb{Z}$ acting on $\mathbb{Q}^2$ failing in characteristic zero for the other reason. Everything in §07 up to that point is over an arbitrary field.

Sections 8 and 9 fix $F = \mathbb{C}$ throughout, where both hypotheses hold at once and neither can fail: $\mathbb{C}$ is algebraically closed, so Schur's lemma gives scalars, and $\operatorname{char} \mathbb{C} = 0$ divides no group order, so the averaging is legal. That is why the complex theory is so much cleaner than the general one. The modular case, where the characteristic does divide $|G|$, is a different subject and the chapter does not enter it.

One further hypothesis is local. Proposition 23.4.6, which says an algebra closed under adjoints acts semisimply, lives on a real or complex inner product space; it is the reason Chapter 12 §07 needed no averaging, and it is used nowhere else.

23.-1.2 What the chapter quotes, and what it does not prove

The chapter imports no analysis. No limit, no continuity, no compactness argument appears in it, and it adds nothing to the list of analytic facts **(A1)–(A6)** that Chapter 16 introduced — Chapter 16's numbering, not §01's algebra axioms (A1)–(A3) — it does not use that list at all. Roots of unity are Chapter 0 §06's complex numbers and nothing more.

Six things are quoted with credit and proved nowhere in the book. Each is flagged again where it appears, and **nothing in the chapter rests on any of them.** No theorem here rests on an exercise either; where an exercise is pointed at, the text says that an exercise is not a proof.

- **Section 6 — the Wedderburn–Artin theorem over a division ring**: that a finite-dimensional semisimple algebra over *any* field is a product $M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$ with each D_i a division algebra. The chapter proves only the algebraically closed case, where every D_i is F. The reason given is the honest one: the book has never built division rings, and the quaternions are one example, not a theory.
- **Section 6 — Wedderburn's little theorem**, that a finite division ring is a field. Same reason. Its effect on the theorem above is recorded and then not used.
- **Section 8 — that the character table does not determine the group.** Two nonisomorphic groups of order 8, the symmetries of a square and the unit quaternions, have identical tables. §08 names them in a warning and constructs neither; the book has neither the group theory to do it nor any need of the fact. What the character table *does* determine — every complex representation of the group up to equivalence — is proved, in Corollary 23.8.8.
- **Section 9 — the structure theorem for finite abelian groups**, that every one is a direct product of cyclic groups. What §09 actually proves is $|\widehat{G}| = |G|$ for every finite abelian G (Proposition 23.9.3), and $\widehat{G} \cong G$ for cyclic groups and for finite direct products of them (Theorem 23.9.4). The general isomorphism is stated as true and not proved, and §09 says which of its later statements would need it: none. The remark points at Chapter 10 §07's Smith normal form as the linear-algebra shadow of the missing argument, and at an exercise there, and adds that the exercise assumes an input it does not prove.

- **Section 3 — the converse of Proposition 23.3.6**, that a finite-dimensional algebra with zero radical is semisimple. Not proved and never used: whenever semisimplicity is needed later it is established directly, by Maschke's theorem or by exhibiting a decomposition.
- **Section 1 — when the commutant of a single matrix is the algebra of polynomials in it.** The exact condition is that the minimal and characteristic polynomials agree. §01 computes two commutants where they do and one where they do not, states the criterion in a remark with credit, and proves nothing about it.

Two further names are mentioned and no statement of theirs is used. §05 records that its proof of the double commutant theorem is the finite-dimensional shadow of Jacobson's density theorem, and that there is an analytic theorem of the same shape, von Neumann's, for operators on a Hilbert space. Neither is proved and neither is needed.

Four neighboring subjects are left outside on purpose, and §10 lists them with the reason in each case: induced representations and Frobenius reciprocity, which need cosets of a subgroup that Chapter 0 does not have; modular representation theory, which is what happens when Maschke's theorem fails; character tables as a computational craft, of which the chapter computes three small ones; and representations of infinite groups, where every proof here loses both of its uses of finiteness.

23.-1.3 What you need

The chapter leans on Chapters 0 to 6 and on Chapters 9, 10, and 12, and takes one or two facts each from Chapters 7, 8, 11, 14, and 15. **Chapters 13 and 16 to 22 are not cited anywhere in it.** No norm, no singular value, no positive semidefinite matrix, no matrix inequality, no convexity, and no geometry appears at any point. Eigenvalues enter only through Chapter 9, and almost always through one fact: over an algebraically closed field, an operator on a nonzero space has one.

Chapter 0 §10 is all the group theory the book has — groups, subgroups, homomorphisms, the symmetric group and cycle notation, and nothing else. There is no Lagrange's theorem, no coset of a subgroup, and no quotient group. Where this chapter needs more than that, it builds it in place and says so: conjugacy classes, and the proof that they partition the group, in §08; the order of an element and the cyclic subgroup it generates, and the direct product of two groups, in §09; and, in an exercise of §01, the class sums that span the center of $F[G]$.

- **Chapter 0** — matrix multiplication and the three views of a product, which carry §01's and §02's computations with matrix units; the characteristic of a field and the arithmetic of integer multiples of 1_F, which is exactly Maschke's hypothesis; the complex numbers and their roots of unity, which are every value a character takes; and equivalence relations and partitions, for the conjugacy classes: Definition 0.9.9, Theorem 0.9.4, Theorem 0.9.2, Definition 0.7.2, Lemma 0.7.4, Theorem 0.8.5, Example (Roots of Unity), Theorem 0.6.4, Definition 0.5.2, Theorem 0.5.2, Theorem 0.2.3, Theorem 0.2.10, Definition 0.10.1, Definition 0.10.2, Definition 0.10.3, Example (Groups), Example (Computing with cycles), Theorem 0.10.1, Theorem 0.10.3, Theorem 0.10.2, Theorem 0.10.4.
- **Chapter 1** — the constant companion: subspaces, spans, bases, and dimension, and above all the direct sum criteria, since every decomposition theorem in the chapter is a statement that some sum is direct. Theorem 1.7.6 supplies the ordinary complement that Maschke's averaging repairs, and Theorem 1.6.8 is how a minimal invariant subspace is recognized as irreducible: Example (The Standard Vector Spaces), Theorem 1.2.1, Theorem 1.2.2, Definition 1.3.2, Theorem 1.3.1, Example (Standard Bases), Theorem 1.6.5, Theorem 1.6.8,

Theorem 1.6.4, Theorem 1.7.6, Theorem 1.7.4, Theorem 1.7.5, Theorem 1.7.3.

- **Chapter 3** — three facts. Permutation matrices are how the symmetric group becomes operators, and the theorem that a one-sided inverse of a square matrix is two-sided is what turns the row orthogonality of characters into column orthogonality in one line: Definition 3.4.2, Lemma 3.4.7, Theorem 3.4.5.
- **Chapter 2** — the workhorse, and the chapter's largest debt after Chapter 1. Kernels and images give every invariance argument, and Rank–Nullity is used in four sections: Theorem 2.1.6, Theorem 2.2.1, Theorem 2.2.2, Theorem 2.2.3, Theorem 2.3.1, Theorem 2.4.2, Theorem 2.5.1, Theorem 2.5.3, Theorem 2.5.4.
- **Chapter 4** — products and quotients of spaces give the quotient algebra, the first isomorphism theorem, and the n-fold copy in the proof of the double commutant theorem; the matrix of a map turns $\mathcal{L}(V)$ into $M_n(F)$ as algebras; and projections and the trace are the whole of §08: Theorem 4.1.4, Theorem 4.1.2, Definition 4.4.1, Definition 4.4.3, Theorem 4.4.1, Theorem 4.4.5, Theorem 4.4.7, Definition 4.5.1, Theorem 4.5.1, Definition 4.5.2, Theorem 4.5.3, Theorem 4.5.4.
- **Chapter 5** — the dual space, in two places. Burnside's proof runs on rank-one operators written as $v\varphi$, so it needs a dual basis and $\dim V^* = \dim V$; and the annihilator and the dual map appear in the exercises of §§02 and 07, which classify the right ideals of $M_n(F)$ and build the contragredient representation: Theorem 5.1.1, Definition 5.1.3, Corollary 5.1.4, Definition 5.2.1, Theorem 5.2.3, Theorem 5.2.2, Corollary 5.3.5, Definition 5.4.1, Theorem 5.4.1, Theorem 5.6.7.
- **Chapter 6** — polynomials. Division with remainder and the minimal polynomial give $\dim F[T] = \deg m_T$; the ideals of $F[x]$ are the commutative model §02 generalizes; unique factorization and the primary decomposition are what §03 needs to identify Chapter 10's semisimple operator with this chapter's; the Fundamental Theorem of Algebra is where "algebraically closed" is grounded; and Lagrange interpolation writes the projections onto the eigenspaces of a diagonalizable operator as polynomials in it, which is how §05 computes a commutant: Theorem 6.1.1, Theorem 6.1.3, Definition 6.2.1, Example (First Ideals), Proposition 6.2.1, Definition 6.3.1, Theorem 6.3.5, Theorem 6.5.1, Theorem 6.6.5, Theorem 6.6.1, Theorem 6.7.3, Theorem 6.7.1.
- **Chapter 7 §03** — one fact, used for one representation: the sign of a permutation is multiplicative, which is what makes the sign representation of S_n a representation: Definition 7.3.2, Theorem 7.3.1. It is named where §07 lists its examples, and checked in one of that section's exercises; no proof here uses it.
- **Chapter 8 §01** — one fact: block diagonal matrices multiply blockwise, which is how §06's worked example recognizes a product of matrix algebras inside $M_n(F)$: Theorem 8.1.4. It is used there and nowhere else, and not inside a proof.
- **Chapter 9** — invariant subspaces, eigenvalues, and the commuting-operator theorems this chapter re-derives and explains. Theorem 9.2.1 and Theorem 9.2.4 are the single fact algebraic closure is spent on; Theorem 9.8.1 and the cyclic machinery settle §03's comparison with Chapter 10; and §§04 and 10 look back at Theorem 9.9.3 and Theorem 9.9.5 from the new height: Definition 9.1.1, Proposition 9.1.2, Example (A Rotation Has No Invariant Lines), Theorem 9.2.1, Theorem 9.2.4, Theorem 9.2.2, Theorem 9.4.1, Theorem 9.5.1, Theorem 9.8.1, Theorem 9.8.2, Corollary 9.8.3, Lemma 9.9.2, Theorem 9.9.1, Theorem 9.9.3, Theorem 9.9.5.
- **Chapter 10** — three separate debts. The cyclic subspace machinery of §05 is what §03 needs to prove that a semisimple operator is one whose generated algebra acts semisimply; the Weyr commutant is re-read in §05 and again in §10 as the case where the double commutant theorem has nothing to say; and the Smith normal form is what §09 points at for the gap it cannot close: Definition 10.5.2, Proposition 10.5.1, Theorem 10.5.2, Theorem 10.5.3, Proposition 10.4.6, Theorem 10.7.3, Exercise (C1: The same algorithm

over the integers) (pointed at, not used), Definition 10.8.1, Proposition 10.8.2.

- **Chapter 11** — the inner product, in three places and never as a standing assumption: §04's proposition that an adjoint-closed algebra acts semisimply, §08's inner product on class functions, and §09's proof that the character matrix is unitary: Definition 11.1.1, Theorem 11.1.4, Definition 11.3.1, Theorem 11.3.2, Definition 11.6.1, Theorem 11.6.1, Theorem 11.7.3.
- **Chapter 12** — the chapter this one explains. Section 9's circulants, the Fourier matrix, the convolution theorem, and the diagonalization of every circulant at once are §09's subject; §07's commuting normal families and §10's disjoint-spectra commutant are reread in §§04 and 10: Corollary 12.4.5, Theorem 12.7.1, Definition 12.9.1, Corollary 12.9.5, Theorem 12.9.1, Theorem 12.9.4, Theorem 12.9.6, Theorem 12.9.3, Theorem 12.9.2, Example (The Fourier matrix of size four), Example (The running example, diagonalized), Corollary 12.10.4.
- **Chapter 14 §12** — the definition of an algebra, which this chapter uses as it stands, and the quaternions, which are the chapter's only noncommutative division algebra and are named as such twice: Definition 14.12.1, Definition 14.12.3. Both are carried over as definitions, in §01's own definition of an algebra homomorphism and in its examples; no proof here cites Chapter 14.
- **Chapter 15** — the free vector space on a set, which is what the group algebra is built from; the fact that a multilinear map may be prescribed freely on bases, which is how the group algebra's multiplication is shown to exist and be unique; and the two-sided ideal and quotient algebra of §10, which §02 takes over without reproving: Theorem 15.1.1, Definition 15.2.3, Lemma 15.2.2, Definition 15.10.3, Lemma 15.10.5.

23.-1.4 Roadmap

- **Associative Algebras.** The vocabulary, and the two examples the rest of the chapter runs on. Subalgebra, with the identity demanded by name and a set exhibited that satisfies every other clause without being one; homomorphism, kernel, and image, with transposition as the nonexample that reverses products; the subalgebra generated by a set, which for one operator is $F[T]$ of dimension $\deg m_T$. Then the group algebra $F[G]$, built as the free vector space on G with the multiplication forced by bilinearity, and commutative exactly when G is abelian. Cayley's theorem for algebras: left multiplication embeds any finite-dimensional algebra in the operators on itself, so matrices are the general case and not a special one. Finally commutants, three of them computed by comparing entries, and the proof over every field that the center of $M_n(F)$ is the scalars.
- **Ideals and Quotients.** In a noncommutative algebra there are three kinds of ideal, and which kind is meant decides every later statement. One product rule does the work — sandwiching a matrix between two matrix units extracts one entry and puts it anywhere — and out of it comes the chapter's first real theorem: over **every** field, $M_n(F)$ has no two-sided ideals but $\{0\}$ and itself. The left ideals are another matter, and they are classified completely, by subspaces of row vectors, with the column ideals minimal among them and the transposed statement for right ideals recorded so the two are not confused. Then quotient algebras and the first isomorphism theorem, with two instances: the group algebra of a cyclic group is $F[x]$ with one relation, and the upper triangular 2×2 matrices modulo their strictly upper part are $F \times F$.
- **Simple and Semisimple Algebras.** A second sense in which something cannot be broken apart — this time a space rather than an algebra — and the section opens by warning that both are spelled "simple". Representation of an algebra, invariant subspace, irreducible space. Then the theorem, over any field: every invariant subspace has an invariant complement if and only if the space is a sum of irreducibles, if and only if it is a

direct sum of finitely many. An algebra semisimple in its own regular representation acts semisimply on every space. The radical collects what no irreducible representation can see; a nilpotent ideal always lies inside it, a semisimple algebra has none, and the upper triangular matrices have a one-dimensional one, computed. Finally the reconciliation with Chapter 10: an operator is semisimple exactly when $F[T]$ acts semisimply — proved with no hypothesis on the field, because the argument never leaves F.

- **Schur's Lemma.** Maps that respect the action, and how few of them there are. Part (a), that a map between irreducibles is zero or an isomorphism, needs nothing of the field and says the commutant of an irreducible action is a division algebra. Part (b), that over an algebraically closed field the division algebra is F, needs an eigenvalue and nothing else. The failure over $\mathbb{R}$ is then exhibited in full, the commutant of the rotation representation computed by multiplying out and identified with $\mathbb{C}$, every nonzero element invertible exactly as part (a) predicts. Two corollaries: an operator commuting with an irreducible family is a scalar, and an irreducible space for a commutative algebra is a line. Then the look back — Chapter 9 §09's common eigenvector was this, its clever trick Schur's lemma unnamed — and a proposition that an adjoint-closed algebra acts semisimply, which recovers Chapter 12 §07 for commuting self-adjoint families out of two general facts. A warning closes it: "the commutant is the scalars" follows from irreducibility and is never a test for it.
- **Burnside and the Double Commutant.** Two theorems about how much of an algebra its commutant records. Burnside: over an algebraically closed field, a subalgebra of $\mathrm{End}(V)$ containing the identity and acting irreducibly is **all** of $\mathrm{End}(V)$ — irreducibility, a statement about subspaces, forces the largest dimension there is. The proof takes a member of least rank, shows that rank is one, and spreads one rank-one operator over the whole space in two directions; a version for a multiplicative set rather than an algebra is recorded alongside it. The double commutant theorem: if A acts semisimply and contains id_V then $A'' = A$, over **any** field. Its engine is matching a single vector, and the upgrade to a whole basis is done by copying the space n times. Both hypotheses are shown necessary and neither is shown necessary in the converse direction. The section ends by re-reading Chapter 10's Weyr commutant and Chapter 12's disjoint-spectra commutant as computations of A', and by doing the two extremes, the diagonal matrices and the scalars.
- **The Structure Theorem.** Products of algebras, and the observation that such a product has dimension $\sum n_i^2$. The regular representation, whose invariant subspaces are the left ideals. Then Wedderburn in the case the book can carry: over an algebraically closed field a nonzero semisimple algebra is a product $M_{n_1}(F) \times \cdots \times M_{n_k}(F)$, and k and the multiset of the n_i are determined, being the number of isomorphism classes of irreducible spaces and their dimensions. The proof splits the regular representation, keeps one representative of each class, computes the commutant of the image with Schur's lemma, and lets the double commutant theorem close the gap; Burnside is **not** used. The corollary reads off $\dim A = \sum n_i^2$ and $\dim Z(A) = k$, which between them pin down the block sizes of a small algebra by arithmetic alone. The general theorem over a division ring and Wedderburn's little theorem get their own subsection, stated with credit and proved nowhere. Two decompositions are computed: $\mathbb{C}[C_4]$ with its four idempotents, and $M_2(\mathbb{C})$, already in Wedderburn form.
- **Group Representations.** A homomorphism $G \to \mathrm{GL}(V)$, with the trivial, sign, permutation, rotation, and regular representations, and a nonexample that fails only by not being well defined on a quotient. Subrepresentations, equivalence, irreducibility, direct sums. Then the bridge: representations of G over F correspond exactly to algebra homomorphisms $F[G] \to \mathrm{End}(V)$, carrying invariant subspaces to invariant subspaces and intertwining maps to $F[G]$-maps, so §§01 to 06 apply verbatim. Maschke's theorem: average a projection over the group and the average is a projection onto the same subspace

that commutes with the action. The hypothesis $\operatorname{char} F \nmid |G|$ is spent on one division, and without it the unnormalized sum annihilates the subspace instead of fixing it; $\mathbb{Z}/p\mathbb{Z}$ on $\mathbb{F}_p^2$ is worked out, and a warning drops finiteness instead and breaks the theorem the other way. Complete reducibility follows by induction, so $F[G]$ is semisimple. The section closes with the two families §08 will compute characters for: all the irreducible complex representations of $\mathbb{Z}/n\mathbb{Z}$, and three for S_3 with every matrix written out.

- **Characters and Orthogonality.** Conjugacy classes are built here, since the book has not met them, and shown to partition the group. A character is one number per group element: constant on classes, additive over direct sums, equal to the dimension at the identity and largest there in absolute value, conjugated by inversion — and **not** a homomorphism. The class functions form an inner product space whose dimension is the number of classes. Two averaging lemmas — the average of a representation is the projection onto the fixed space, and the trace of $f \mapsto BfA$ is $(\operatorname{tr} A)(\operatorname{tr} B)$ — combine into a count of intertwining maps, and Schur's lemma turns that count into a 0 or a 1: the irreducible characters are orthonormal. The criterion $\langle \chi, \chi \rangle = 1$ then decides irreducibility without looking at a vector, and decomposes the permutation representation of S_3 in three lines. The deepest result is that the characters **span** the class functions, by a third averaging argument run on the regular representation; so the number of irreducible representations is the number of conjugacy classes. Out of it come the sum of the squares of the degrees — the identity §06 proved by another route entirely — and column orthogonality, from a one-sided inverse. The tables of S_3 and $\mathbb{Z}/4\mathbb{Z}$ are computed and every relation checked.
- **Finite Abelian Groups and the Fourier Transform.** The order of an element, the cyclic group, and the direct product of groups are built first, Chapter 0 having none of them. Over $\mathbb{C}$ an abelian group has only one-dimensional representations, so a character is a homomorphism $G \to \mathbb{C}^\times$ whose values are roots of unity, and the characters form a group $\widehat{G}$ under pointwise multiplication. It has the same order as G, always; it is isomorphic to G for cyclic groups and for products of them, and the section stops there and says exactly what it is not proving. Lining G up against $\widehat{G}$ gives a square character matrix, which is unitary, and inversion and Plancherel are what a unitary matrix always gives. Then the three identifications: for $\mathbb{Z}/n\mathbb{Z}$ the character matrix **is** the Fourier matrix; the group algebra of any finite abelian group splits into the functions on $\widehat{G}$, which for the cyclic group is the convolution theorem; and the circulants are the group algebra of $\mathbb{Z}/n\mathbb{Z}$ acting by multiplication, with the basis reversal that Chapter 12's indexing forces spelled out in a warning. The case $n = 4$ is checked against Chapter 12's numbers.
- **What the Chapter Buys.** A closing section that introduces nothing. Four moves are named and located; four earlier theorems are read again from here — simultaneous diagonalization, commuting normal families, circulants and the discrete Fourier transform, and the two commutant computations of Chapters 10 and 12. A table gives, for each main theorem, the hypothesis on the field it needs and the counterexample that shows the hypothesis is not decoration, with three comments on what the table reveals. Then the list of what was quoted, the list of what was left outside and why, and a last look back at what a clean answer cost.

23.-1.5 Named moves

- **Average over the group to build an invariant object.** Start with something that depends on a choice — a projection, an arbitrary linear map, a vector — and replace it by the average of its translates. The average is invariant because translating permutes the summands. This is the whole of Maschke's theorem, the whole of the orthogonality relations, and the whole of the proof that the characters span. It costs one division by $|G|$, and that division is the chapter's only hypothesis on the characteristic.

- **Use Schur's lemma to force a scalar, then take a trace to find which one.** Build an operator by some construction, check that it commutes with everything in sight, conclude it is λ id, and compute λ from the trace. Two lines that replace an unknown operator by a number, over and over, and the exact price is that F be algebraically closed.
- **Sandwich with matrix units.** The identity $\mathbf{E}_{ij}\mathbf{X}\mathbf{E}_{kl} = x_{jk}\mathbf{E}_{il}$ extracts one entry of an arbitrary matrix and puts it wherever you like. An ideal that contains anything nonzero therefore contains every matrix unit, which is the simplicity of $M_n(F)$ in three lines and over any field.
- **Let the algebra act on itself.** An abstract algebra has no vectors until you give it some, and the cheapest supply is the algebra. Left multiplication makes it an algebra of operators (which is why matrices are general), and the invariant subspaces of that action are the left ideals — so a question about ideals and a question about representations are one question.
- **Copy the space to upgrade one vector to a basis.** A density argument that matches an operator on a single vector becomes a proof that two operators are equal, once it is run on $V \oplus \cdots \oplus V$ with the basis of V packed into one vector of the copy. The only work is checking that the copied algebra inherits the hypotheses, and semisimplicity is the hypothesis that survives copying.
- **Count with $\sum n_i^2$.** The dimension identity of the structure theorem, read for a group algebra as $|G| = \sum_i (\dim V_i)^2$, pins down the degrees of all the irreducible representations of a small group from the number of conjugacy classes alone, before a single matrix is written. It does for this chapter what Rank–Nullity does for Chapter 2.
- **Read a family of operators as the algebra it generates.** A commuting family generates a commutative algebra; a group of invertible operators spans one; a single operator generates $F[T]$. Invariant subspaces, commutants, and irreducibility are unchanged by the passage, so a theorem about algebras is a theorem about every family the book has studied. That is what lets the last section re-prove four earlier results without a new idea in any of them.